HARSH SARAWAGI

Final Year Undergraduate
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Indian Institute of Technology Kanpur

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Examination	Degree/Certificate	Institute	Year	CPI/%
Graduation	B.Tech	Indian Institute of Technology Kanpur	2027	**7.7**/10.0
Intermediate/+2	CBSE(XII)	Mahatma Hansraj Modern School, Jhansi	2023	91.2%
Matriculation	CBSE(X)	Jai Academy, Jhansi	2021	97.8%

Scholastic Achievements

- Qualified for the National Talent Search Examination (NTSE), ranking among the **top 0.2% of 1 Million+** candidates. (*2021*)

Work Experience

Software Developer Intern
May'26 – July'26

BNY

- Designed and developed a unified **Health Check Agent** as a **Python/FastAPI** backend implementing **MCP with JSON-RPC 2.0**, exposing **19+ tools** for application monitoring and operational support, through **MCP clients and Swagger UI** for seamless tool interaction.
- Integrated **Oracle DB, Splunk and App Engine** to provide centralized application health, log and operational monitoring across **production and QA environments**, along with a separate **RAG-based tool** for retrieving application-specific information and technical documentation.
- Built a **parallel hourly-scheduled monitoring pipeline** within the same server to execute automated health checks, identify application anomalies and trigger **severity-based email alerts** to the team, enabling proactive issue detection.
- Integrated the MCP server with BNY's internal AI platform using **Nuggets**, enabling an LLM-powered workflow to interpret **natural-language requests**, dynamically select the appropriate monitoring tools, and directly deliver results through the **chatbot interface**.

Key Projects

Predator–Prey RL Simulator
Mar'26 – Apr'26

Course Project under Prof. Washim Mondal

- Built a **predator-prey grid-world simulator** with state/action modeling, stochastic transitions and reward computation.
- Implemented **Value Iteration, Policy Iteration and neural-network-based Policy Gradient** algorithms using **PyTorch** and **SciPy sparse matrices** for efficient computation and memory usage.
- Benchmarked **runtime, convergence, and scalability** across increasing state-space sizes and compared optimization strategies.
- Improved **policy stability** using **Adam**, achieving ∼13+ reward per episode versus ∼11 with **Stochastic Gradient Ascent**, with smoother and faster convergence.

AI-Assistant for Programming
Dec'24 – Jan'25

Google Development Group, IIT Kanpur

- Built an **AI-powered competitive programming platform** for performance analytics, rating prediction and practice.
- Developed a **Chrome Extension** providing real-time insights, rating predictions and personalized recommendations directly within the Codeforces platform for targeted practice.
- Designed a **scalable PostgreSQL-backed system** for managing user profiles, submissions, contests and performance data across the platform.
- Developed **RESTful APIs with secure authentication** and deployed the platform using Docker and Gunicorn on Oracle Cloud for reliable application hosting.

STUNT Reimplementation
Aug'26

Self-Project

- Reimplemented **STUNT** from scratch for self-supervised tabular representation learning with perturbation-based episodic ProtoNet training and few-shot classification.
- Built a complete end-to-end pipeline for **preprocessing, training, pseudo-validation, checkpointing and few-shot evaluation** across multiple datasets.
- Achieved **+6.2 pp accuracy and +10.7 pp F1** over raw features at 1-shot on Income census across 100 **independent evaluation trials**, demonstrating consistent low-shot gains.
- Evaluated **1/2/5/10-shot performance** across WDBC and Income, analyzing low-shot gains and **variance** over raw features.

Illuminating Language
May'24 – Jul'24

The Brain and Cognitive Science Club, IITK

- Built an **IITK-specific RAG chatbot** to answer user queries using **institute-specific information** effectively.
- Scraped **IITK data** and implemented **semantic search** with **InstructorEmbeddings** and **ChromaDB** for efficient and accurate context retrieval from institutional resources.
- Improved **retrieval accuracy** using **RAG Fusion, Query Rewriting, Conversation History and Query Routing**, while leveraging **system prompting** for more relevant response.
- Developed an interactive **Streamlit interface** for delivering **context-aware responses** to institutional queries seamlessly.

Full-Stack Social Media Web App
Jun'24

Self-Project

- Built a social media app with **user authentication, friend requests, post sharing, likes, comments and user interactions**, supporting core social networking functionality.
- Developed **RESTful backend APIs** using **Node.js** and **Express**, with **MongoDB** and **Mongoose** for schema management and persistent data storage.
- Implemented a responsive **React.js frontend** with **Zustand** for efficient state management across application components and user interactions.
- Structured the application with **independently deployable client and server services**, simplifying deployment, maintenance and application management.

Technical Skills

- **Programming Languages:** C, C++, Python, JavaScript, SQL
- **Libraries & Utilities:** NumPy, Pandas, PyTorch, scikit-learn, Mongoose, LangChain, BeautifulSoup, Selenium
- **Frameworks & Technologies:** React, Express, FastAPI, Django, MongoDB, Streamlit, MCP, JSON-RPC, HTML, CSS

Positions Of Responsiblity

Academic Mentor
Jul'24 – May'25

Center for Mental Health and Wellbeing, IIT Kanpur

- Provided academic mentorship to first-year students in Engineering Drawing (TA111) through regular doubt sessions.
- Assisted the Institute Counselling Service(ICS) in conducting a 10-day orientation program for over 1200 newly admitted students, helping them transition smoothly into campus life.

Relevant Courses

Data Structures & Algorithms	Introduction to RL	ML for Signal Processing	Probability and Statistics
Introduction to Machine Learning	Fundamentals of Computing	Blockchain Technology & Application	Foundation of modern AI